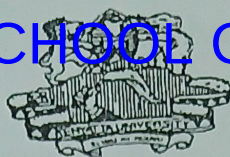


SCH 111 CURTESY OF CALVINE ABABU SCHOOL OF EDUCATION



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

SCH 111: INTRODUCTION TO PHYSICAL CHEMISTRY

2ND SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS

- Answer ALL the questions
- Show your working clearly, be brief and to the point
- Additional information is provided below

$$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.08205 \text{ l-atm K}^{-1} \text{ mol}^{-1}, 0^\circ\text{C} = 273 \text{ K}, N = 6.022 \times 10^{23} \text{ mol}^{-1}$$

Question 1 (20 marks)

- State four assumptions on which the kinetic theory of ideal gas is based.
 - The expression $pV = \frac{1}{3}nm\overline{u^2}$ was obtained from the kinetic theory. What do the symbols n , m and $\overline{u^2}$ stand for?
 - Why do real gases deviate from ideal behaviour?

- State Dalton's law of partial pressures.

- 7g NO 16g O₂ 7/23

Seven grams of nitrogen gas and sixteen grams of oxygen gas are introduced into an evacuated vessel of 50 litre capacity at 50°C. Determine the total pressure of the mixture assuming ideal behaviour (N=14, O=16). 16/23

- Calculate the pressure exerted by 20 g of nitrogen gas enclosed in a 1 litre vessel at 25°C. Assume that nitrogen behaves ideally. $PV = nRT$

Mass
Volume

$$PA = \frac{m}{V} \times P_f$$

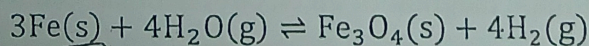
$$P = \frac{nRT}{V}$$

- What is the density of nitrogen gas in the above cylinder?
 - A deep breath of air has a volume of 1.05 litres at a pressure of 1 atm and a body temperature of 37°C. Calculate the number of molecules in the breath.

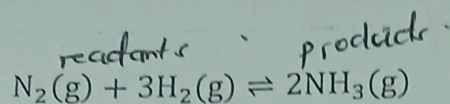
$$\frac{PV}{RT} = \frac{nRT}{RT} \quad n = \frac{PV}{RT}$$

Question 2 (20 marks)

- Write the equilibrium expression K_c for the reactions below



Products
Reactants



- (b) The equilibrium constant K_c for the reaction $\frac{1}{2}\text{H}_2(\text{g}) + \frac{1}{2}\text{I}_2(\text{g}) \rightleftharpoons \text{HI}(\text{g})$ at 718 K is 7.07

(i) What is the value of K_c at 718 for the reaction $\text{HI}(\text{g}) \rightleftharpoons \frac{1}{2}\text{H}_2(\text{g}) + \frac{1}{2}\text{I}_2(\text{g})$

(ii) What is the value of K_c at 718 for the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$

(iii) The equilibrium constant is 4.00 for the reaction $K_c = 4$
 $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$
 at 100°C. Calculate the number of moles of each substance present at equilibrium if the starting number of moles are 4, 4, 0, 0 for acid, alcohol, ester and water respectively.

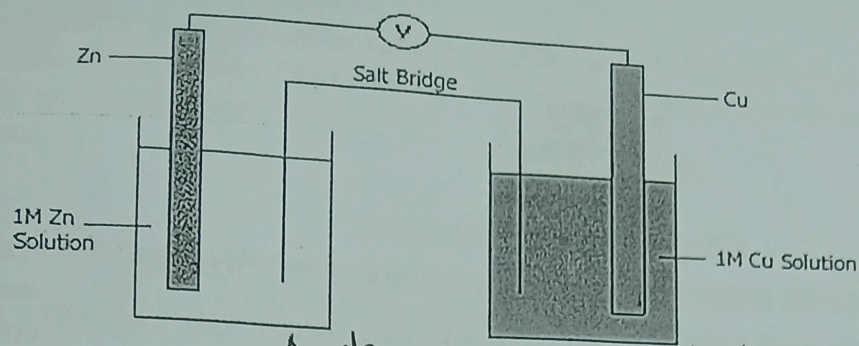
- (c) (i) The pH of household bleach is approximately 12 at 25°C. What is the hydrogen concentration of the bleach.
 (ii) Propanoic acid is a weak acid whose ionization constant is $K_a = 1.336 \times 10^{-5}$. What is the pH of 0.01 M $\text{C}_2\text{H}_5\text{COOH}$ solution?

- (d) (i) What is a buffer solution?
 (ii) A buffer solution was analysed and found to be 0.183 M in acid and 0.625 M in salt. Its pH was measured and found to be 5.82, calculate the dissociation constant K_a for the acid.

Question 3 (20 marks)

- (a) (i) State Raoult's law
 (ii) Draw vapour pressure – composition diagram for a two component solution showing negative deviation from Raoult's law.
 (iii) You have a solution made up of 1.0 mol of A and 4.0 mol of liquid B. The vapour pressure of pure A is 80 mmHg and that of B is 20 mmHg. Assuming the solution is ideal, calculate the vapour pressure of A and B above the solution and calculate the composition of the vapour in mole fractions

- (b) The diagram shows a schematic diagram of an electrochemical cell using beakers for each half cell at 25°C.

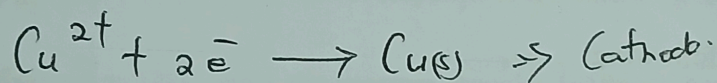
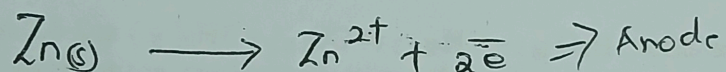


Given also that $E^\circ(\text{Zn} | \text{Zn}^{2+}) = -0.76 \text{ V}$ and $E^\circ(\text{Cu} | \text{Cu}^{2+}) = +0.34 \text{ V}$ at 25°C

- Write the cell in the shorthand notation using lines
- Write down the anode reaction
- Write down the cathode reaction
- Write down the overall cell reaction
- Calculate the E° for the cell at 25°C
- What would be the cell potential if the $[\text{Zn}^{2+}] = 0.5 \text{ M}$ and $[\text{Cu}^{2+}] = 2 \text{ M}$

Anode $\Rightarrow$ Oxidation $\rightarrow$ losing electrons

Cathode $\Rightarrow$ Reduction $\rightarrow$ gaining electrons.



SCH 111 CURTESY OF
CALVINE ABABU SCHOOL OF
EDUCATION



KENYATTA UNIVERSITY
CONTINUING EDUCATION PROGRAMME (CEP) 2024/2025
EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION
SCH 101/111: INTRODUCTION TO PHYSICAL CHEMISTRY

AUGUST 2024 SESSION

TIME: 2 HOURS

INSTRUCTIONS:

Answer question **ONE** (30 marks) and any other **TWO**

Some useful constants

$R = 8.314 \text{ J Mol}^{-1} \text{ K}^{-1}$, $0^\circ\text{C} = 273.15\text{K}$, $F = 96500 \text{ C/Mol}$, $1 \text{ atm} = 760 \text{ mmHg} = 101325 \text{ Nm}^2$

Question 1

- a. At 1400K , K_p for the reaction: $\text{H}_2\text{O}(\text{g}) + \text{CH}_4(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$ is 2.62 atm . Determine the value of K_c for the reaction, giving its dimensions, at the given temperature [4 marks]
- b. Give statements of the following:
 - (a) Gay-Lussac's law [2 marks]
 - (b) Graham's law of diffusion [2 marks]
 - (c) Dalton's law of partial pressures [2 marks]
 - (d) Le Chatelier's principles [3 marks]
- c. A quantity of N_2 gas at 4.6 atm pressure in a 1 litre vessel at 25°C is connected, through a valve, to an evacuated 5 litre vessel. Determine the final pressure after the valve is opened and the gas allowed to equilibrate at the same temperature. [4 marks]
- d. Two moles of helium gas occupy a volume of 30 litres at 25°C . Determine the ideal pressure of the gas where $R = 0.0821 \text{ (L} \cdot \text{atm/mol} \cdot \text{K)}$ [2 marks]
- e. State the Le Chatelier's principle [2 marks]
- f. For the equilibrium: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $K_c = 2.46 \times 10^2 \text{ litre mol}^{-1}$ at 1000K
 - i. Determine K_p at the same temperature [2 marks]
 - ii. Explain the effect of pressure increase on the equilibrium concentrations of the constituent substances [3 marks]

- g. The solubility product of silver chloride, AgCl(s) at 25°C is $2.8 \times 10^{-10} \text{ mol}^2 \text{ litre}^{-2}$. Determine the number of moles of Ag^+ ions present in 400 cm^3 of the saturated solution at the same temperature [4 marks]

Question 2

- What is a Buffer and how does it work? [4 marks]
- Determine $[\text{H}^+]$ of a solution with a pH value of 8.0 [2 marks]
- Determine $[\text{OH}^-]$ of a solution with a pH value of 3 [2 marks]
- Determine the pH of a 0.3M aqueous solution of a weak monoprotic acid with $K_a = 1.8 \times 10^{-4}$ [2 marks]
- Given that solubility for AgCl is $1.67 \times 10^{-5} \text{ mol l}^{-1}$ at 25°C . What is the solubility product of AgCl in 1 liter of water? [2 marks]
- K_{sp} for CaF_2 is $1.7 \times 10^{-10} \text{ M}^2$ at 25°C . What is the solubility of CaF_2 in 0.1 M $\text{Ca}(\text{NO}_3)_2$. [2 marks]
- The molal elevation constant for water is $0.513^\circ\text{C kg mol}^{-1}$. When 0.2mole of sugar is dissolved in 250g of water, calculate the temperature at which the solution boils under atmospheric pressure. [3 marks]
- Calculate the root mean velocity of nitrogen molecules at 25°C given that the RFM for nitrogen is 14.007 [3 marks]

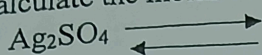
Question 3

- Give the four postulates pertinent to the development of the kinetic theory of ideal gases (4 marks)
- 13.09g of an ideal gas at 298K and 1 atm pressure occupies $1 \times 10^{-2} \text{ m}^3$. Determine the formula mass of the gas (4 marks)
- State Graham's law of diffusion (2 marks)
 - State Dalton's law of partial pressures (2 marks)
- 7g of an ideal gas with formula mass of 32 g mole^{-1} occupies a volume of 10 litres at 30°C . Calculate the pressure of the gas in atmospheres (4 marks)
- Two moles of ideal gas helium occupy a volume of 30 litres at 25°C . Determine the pressure of the gas, using the van der Waals equation
($a = 0.034 \text{ litre}^2 \text{ atm mol}^{-2}$, $b = 2.36 \times 10^{-2} \text{ litre mol}^{-1}$) (4 marks)

Question 4

- Given that a mixture of benzene and toluene boils at 97°C and 1 atm pressure and vapour pressure of pure benzene and toluene are 1.67 0.84 atm respectively. Calculate

- i. The mole fraction of benzene in the mixtures [3 marks]
- ii. The composition of the initial condensate formed on distilling the mixture [3 marks]
- iii. Explain with the help of suitable diagram why a solvent that has dispersed a solute boils at a higher temperature than the pure solvent [3 marks]
- b. The vapor pressure of an aqueous solution is found to be 24.90 mmHg at 25° C. What is the mole fraction of solute in this solution? The vapor pressure of water is 25.756 mmHg at 25° C [4 marks]
- c. Explain the origin of deviation shown by real gas from relationship $\frac{PV}{RT} = 1$ [4 marks]
- d. Calculate the molar solubility of Ag_2SO_4 in water given that $2\text{Ag}^+_{(\text{aq})} + \text{SO}_4^{2-}_{(\text{aq})}$ and $K_{\text{sp}} = 1.4 \times 10^{-5}$ [4 marks]



SCH 111 CURTESY OF
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KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE AND BACHELOR OF
EDUCATION (SCIENCE)

SCH 101 : INTRODUCTION TO PHYSICAL CHEMISTRY

2ND SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS: ANSWER ALL QUESTIONS.

Useful information:

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$= 0.08206 \text{ litre} - \text{atm mol}^{-1} \text{ K}^{-1}$$

$$1 \text{ atm} = 760 \text{ mmHg}$$

$$= 101,325 \text{ Nm}^{-2}$$

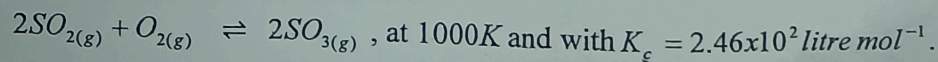
$$0^{\circ} = 273.15 \text{ K}$$

QUESTION ONE: (17 MARKS)

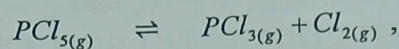
- (a) Explain the *concept of an ideal gas* [3 marks]
- (b) Give *mathematical expressions* of the following laws, defining the terms used:
- (i) Gay-Lussac's law [3 marks]
- (ii) - Graham's law of diffusion [3 marks]
- (c) A gas mixture consists of 0.2 mole of argon and 0.3 mole of helium. If the total pressure of the gas mixture is 0.75 atm, determine the *partial pressures* of the constituent gases [4 marks]
- (d) At 25°C and 0.5 atm pressure, an ideal gas with formular mass 28.0134 g mol⁻¹ occupies a volume of 3 litres. Determine the *mass* of the gas, in *kilograms*. [4 marks]

QUESTION TWO: (16 MARKS)

- (a) State the *principle* that deals, *qualitatively*, with the effects that changes in conditions have on the positions of chemical equilibria [3 marks]
- (b) Consider the following equilibrium:



- (i) Explain the effect of pressure changes on the equilibrium concentrations of the constituent substances [3 marks]
- (ii) Determine the value of K_p at the same temperature [4 marks]
- (c) For the equilibrium reaction:



the fraction of the $PCl_{5(g)}$ molecules that dissociate to reach equilibrium is α .

Derive an expression for K_p in terms of α and the total pressure P [6 marks]

QUESTION THREE: (15 MARKS)

- (a) (i) Determine $[H^+]$ of a solution with a pH value of 8 [3 marks]
- (ii) Determine $[OH^-]$ of a solution with a pH value of 3 [3 marks]
- (iii) Determine the pH of a 0.3M aqueous solution of a weak monoprotic acid with $K_a = 1.8 \times 10^{-4}$ [3 marks]
- (b) Explain the term *buffer solution* [3 marks]
- (c) An aqueous solution of ammonium chloride, NH_4Cl , turns litmus purple. Explain, using relevant chemical equations [3 marks]

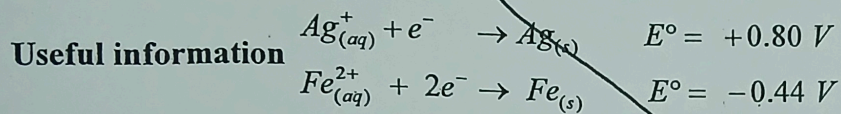
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QUESTION FOUR: (10 MARKS)

- (a)
- (i) What do you understand by the term vapour pressure? [2 marks]
- (ii) Calculate the vapor pressure of a mixture containing 252 g of n-pentane ($M_w = 72$) and 1400 g of n-eptane ($M_w = 100$) at 20°C. The vapor pressure of n-pentane and n-eptane are 420 mm Hg and 36 mm Hg respectively. [4 marks]
- (b)
- (i) Will the mixture boil at higher or lower vapour pressure of each of the respective pure solvents? [2 marks]
- (iii) Give reasons for your answer in question 4b(i) above [2 marks]

QUESTION FIVE: (12 MARKS)

- a) Differentiate between a voltaic and electrolytic cells. [2 marks]
- b)
- (i) Draw labelled diagram of a Standard Hydrogen Electrode (SHE) [3 marks]
- (ii) Write a half-cell equation for the reaction in the half cell above. [2 marks]
- c)
- (i) Given the equation $2Ag_{(s)} + Fe_{(aq)}^{2+} \rightarrow 2Ag_{(aq)}^{+} + Fe_{(s)}$ Calculate the equilibrium constant for the following the reaction at $25^{\circ}C$ given that $E_{cell}^{\circ} = 0.0592 V$ [3 marks]



- (i) Based on the equilibrium constant value, is the reaction by a *spontaneous* chemical reaction or not? [1 marks]
- (iii) Give reasons for your answer [1 marks]

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KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

SCH 111: INTRODUCTION TO PHYSICAL CHEMISTRY

2ND SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS

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Question 1 (20 marks)

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 - The expression $pV = \frac{1}{3}nm\bar{u}^2$ was obtained from the kinetic theory. What do the symbols n , m and $\bar{u}^2$ stand for?
 - Why do real gases deviate from ideal behaviour?

- State Dalton's law of partial pressures.

(ii) ^{7g} ~~Seven~~ grams of ^{NO} ~~nitrogen~~ gas and ^{16g} ~~sixteen~~ grams of ^{O₂} ~~oxygen~~ gas are introduced into an evacuated vessel of ⁵⁰ ~~50~~ litre capacity at ⁵⁰ ~~50~~ °C. Determine the total pressure of the mixture assuming ideal behaviour (N=14, O=16). 7/23
16/23

- Calculate the pressure exerted by 20 g of nitrogen gas enclosed in a 1 litre vessel at 25°C. Assume that nitrogen behaves ideally. $\frac{P_1V_1 = nRT}{V} = \frac{nRT}{V}$

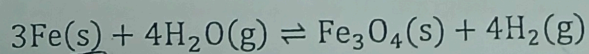
Mass
Volume

- What is the density of nitrogen gas in the above cylinder?
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$$\frac{PV}{RT} = \frac{nRT}{RT} \quad n = \frac{PV}{RT}$$

Question 2 (20 marks)

- Write the equilibrium expression K_c for the reactions below



Products
Reactants